Technical Document #971: Data Engineering - Comprehensive Overview

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Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Data lineage tracks the origin and movement of data through systems. It is essential for debugging, compliance, and impact analysis.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- Data lakes store raw data in its native format.
- Data lineage tracks the origin and movement of data through systems.
- Schema evolution handles changes in data structure over time.